



Semi-supervised concept drift detection and adaptation based on conformal martingale framework

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ABSTRACT

In the realm of industrial applications for machine learning, multiple challenges are frequently encountered, such as concept drift (CD) and the prohibitive costs associated with data labeling. CD refers to the scenario where the underlying data distribution of the model shifts over time, potentially deteriorating model performance. Addressing these challenges, this paper proposes an innovative semi-supervised CD detection method, specifically designed to tackle both CD and the high costs of data labeling in regression tasks. Initially, considering the high expense of acquiring labeled data in industrial application scenarios, a semi-supervised learning strategy based on self-training is utilized. In this strategy, prediction intervals generated by Conformal Prediction (CP) are used to select high-reliability pseudo-labels. Furthermore, to effectively address CD in real-world industrial settings, the Conformal Martingale (CM) is employed for real-time detection. This framework detects changes by identifying increases in martingale values when CD occurs. Upon detection, the model is promptly retrained using the most recent data following the drift. Finally, the proposed method is validated through experiments conducted on three datasets: the UCI dataset, the alumina evaporation process dataset, and the blast furnace ironmaking dataset. Experimental results demonstrate that the proposed semi-supervised method significantly enhances the performance of the original training model. The detection method accurately identifies CD and notably reduces test errors through model retraining, thereby improving the effectiveness of the model in real-world industrial applications.

1. Introduction

As the process of industrial informatization deepens, the massive data streams generated by sensors present unprecedented challenges to traditional machine learning tasks. In response, a variety of methods for analyzing and mining data streams have emerged. However, a notable challenge in online environments is Concept Drift (CD), defined as the evolution of the statistical properties of data over time in dynamic settings. The occurrence of CD can degrade model performance, as the patterns learned during training may no longer apply to new data [1–4]. In industrial settings, CD can result from various factors, including the optimization and adjustment of industrial processes, wear and aging of machinery and equipment, changes in environmental conditions, and new policy implementations. These factors significantly impact models deployed in industrial applications [5]. As illustrated in Fig. 1, effectively detecting and addressing CD not only maintains optimal model performance but also significantly mitigates its potential adverse effects.

Currently, the field of CD detection has become a focal point among

researchers, with detection strategies primarily divided into three categories: methods based on statistical testing, model performance monitoring, and data distribution analysis. Statistical test-based methods identify CD by analyzing the statistical properties of data streams. For example, Page et al. [6] proposed the Page-Hinkley test, which relies on the cumulative difference between the observed value and the current mean value as the detection metric. Methods based on model performance monitoring focus on tracking changes in model performance over time to detect CD. Among these, the Drift Detection Method (DDM) [7] is widely used, issuing drifts or warning signals when the overall error rate of the model significantly increases beyond a specific threshold. Other similar methods include the Early Drift Detection Method (EDDM) [8], EWMA for Concept Drift Detection (ECDD) [9],

and the Hoeffding Inequality-based Drift Detection Method (HDDM) [10]. Finally, data distribution-based methods analyze the differences between historical and new data windows, quantifying the dissimilarities using metrics such as distance. For instance, the ADWIN algorithm

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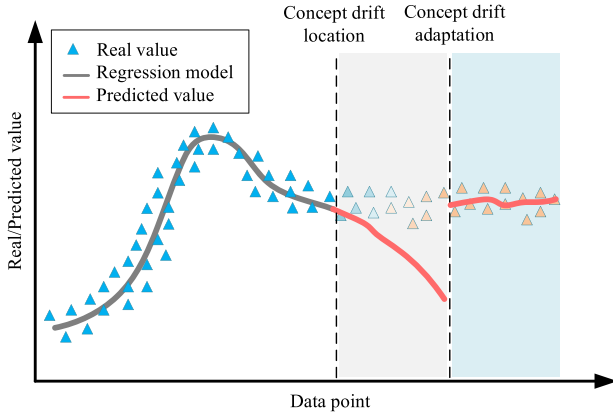


Fig. 1. Concept drift detection and adaptation in regression tasks.

[11] detects drift by comparing mean values between two sequential sub-windows within a variable-size sliding window, identifying drift when the difference exceeds a preset threshold. Bu et al. [12] proposed a change detection method based on the variation in the probability density function between two non-overlapping windows. Tanha et al. [13] introduced a method based on the Conformal Prediction (CP) framework, which detects CD in classification problems across two non-overlapping windows and addresses the issue of high labeling costs. However, despite the extensive range of detection strategies explored in existing research, most focus on classification problems, leaving the detection of CD in regression problems relatively underexplored.

In classification problems, the method based on the CP framework is preferred for its flexibility in calculating non-conformity scores, and it serves as an effective tool for detecting CD in regression tasks. This approach was initially proposed by Vovk et al. [14], with its core based on CP and Conformal Martingale (CM) to perform exchangeability testing. Furthermore, Ho et al. [15] introduced a universal martingale framework employing exchangeability tests to monitor changes in data streams, thereby demonstrating the effectiveness of this approach in practical applications. Specifically, this framework detects CD by calculating the output martingale values of newly received data points to determine whether the dataset satisfies the exchangeability condition. In the absence of CD, the CM values should behave as a bounded

stochastic process; conversely, if CD occurs, the CM values will significantly deviate from the established bounds, indicating a change in model performance. Additionally, Eliades et al. [16] introduced the Inductive Conformal Martingales (ICM) framework for classification tasks, further extending the method's applications. Compared with traditional methods, the CM framework-based approach makes no assumptions about data distribution, adapts to a wide range of real-world application scenarios, and its sensitivity to new data and real-time capabilities ensure its effectiveness in dynamic and online environments.

Another significant challenge associated with large data streams is the prohibitive cost of data labeling, which restricts the direct application of traditional supervised learning models. Although supervised detection methods excel in accuracy, they depend heavily on real-time and precise data labeling, which is challenging to achieve in many practical scenarios. In contrast, unsupervised detection methods are constrained by their high false alarm rates [17]. Given the high cost of data labeling, semi-supervised learning provides an effective solution. This approach reduces reliance on labeled data by allowing the model to utilize unlabeled data. Among the semi-supervised learning methods, the self-training strategy is an effective way to address the issue of high data labeling costs. As illustrated in Fig. 2, the initial phase involves using a small set of labeled data to train the model, which is subsequently used to predict the labels of unlabeled data to generate pseudo-labels. Moreover, high-reliability pseudo-labeled samples are selected for inclusion in the training set for retraining the model [18]. In classification tasks, selecting high-reliability samples is relatively intuitive by assessing the output probability values. However, selecting high-reliability samples for regression tasks becomes more challenging due to the lack of clearly defined criteria. At this juncture, the CP framework can provide variable-width prediction intervals for different test samples by setting the form of non-conformity scores calculation. By selecting samples with narrower prediction intervals to join the high-reliability sample set at a specified confidence level, the CP framework offers an effective method for selecting high-reliability samples in regression tasks. This approach eliminates additional computational burdens, allowing this study to utilize a unified and efficient framework.

To address these issues, this paper proposes a novel semi-supervised CD detection method based on the CM framework, specifically tailored for regression tasks in real-world applications. The main contributions of this paper are summarized as follows:

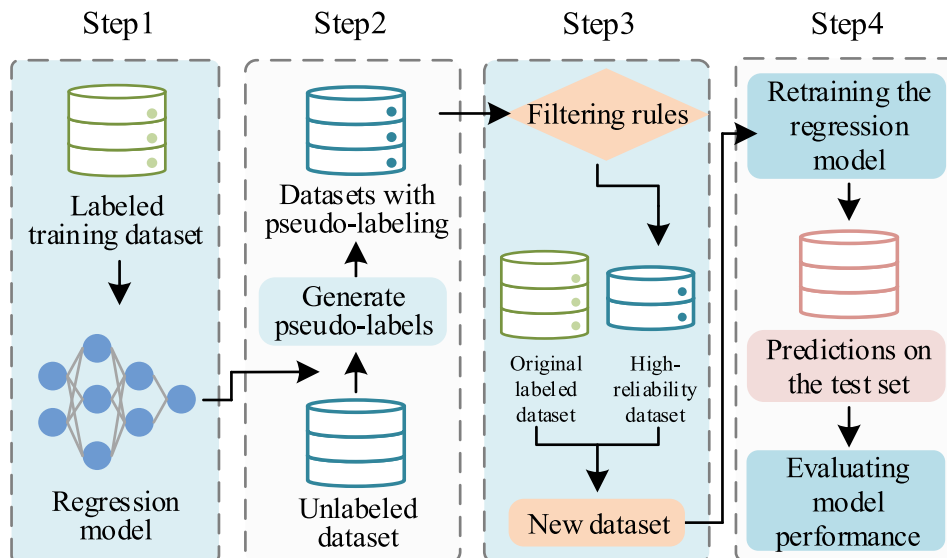


Fig. 2. Self-training strategy steps.

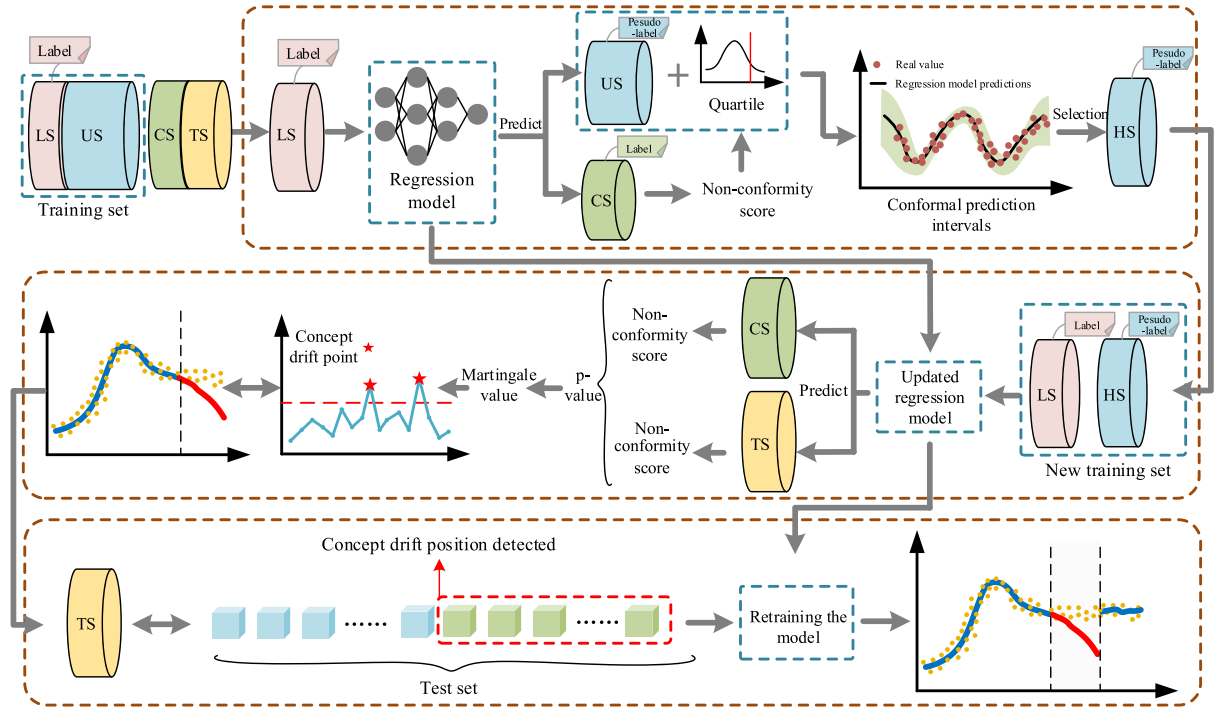


Fig. 3. Overall process of semi-supervised concept drift detection based on conformal martingale framework (where LS denotes the labeled dataset, US denotes the unlabeled dataset, CS denotes the calibration set, TS denotes the test set, and HS denotes the high-reliability set).

- 1) A semi-supervised learning strategy based on the CP framework is proposed. This strategy utilizes the narrow prediction intervals provided by CP to select high-reliability samples for inclusion in the training set, thereby minimizing the influence of pseudo-labeled samples with high uncertainty on the model training process and effectively enhancing the accuracy of the model;
- 2) A CD detection method based on the CM framework is introduced. The method is suitable for dynamic data streams or online environments, and is able to detect new data continuously and adjust the model upon drift detection, thereby avoiding the need for frequent and indiscriminate model retraining. In addition, the method does not necessitate a sliding window on the data streams, which avoids the potential effects related to window length;
- 3) The method presented in this paper offers a generalized framework that does not presuppose any specific statistical distribution of data, making it universally applicable across a diverse range of practical application scenarios. The efficacy and practicality of the proposed method have been validated through experiments conducted on the UCI public dataset, the alumina evaporation process dataset, and the blast furnace ironmaking dataset.

The remaining structure of this paper is organized as follows: Section II presents the background and related work, Section III outlines the proposed algorithm, Section IV details the experimental validation and analysis, and Section V offers the conclusion.

2. Background and related work

2.1. Concept drift

In the context of dynamic data streams, CD refers to the phenomenon where the joint probability distribution of data changes over time [1]. Specifically, consider a sequence of data points in a data stream, $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$, following a certain deterministic distribution F_n . If at a sample point (X_{n+1}, Y_{n+1}) , the original distribution changes to a new distribution, that is:

$$F_n \neq F_{n+1} \quad (1)$$

then it indicates that CD has occurred at the sample point (X_{n+1}, Y_{n+1}) .

2.2. Conformal prediction

CP is an efficient statistical method designed to provide reliable prediction sets for forecasting models. The core idea is that for a dataset $\{D, (X_{n+1}, Y_{n+1})\}$, $D = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ that satisfies exchangeability, a prediction interval is constructed at the test points (X_{n+1}, Y_{n+1}) with a predetermined confidence level $1 - \alpha$, ensuring that the probability of the true target value falling within this interval is at least $1 - \alpha$. Here, exchangeability means that for a given sequence of random variables $\{X_1, X_2, \dots, X_n\}$, the joint distribution remains invariant for any permutation of the indices $\{s_1, s_2, \dots, s_n\}$, that is:

$$F_X(x_1, x_2, \dots, x_n) = F_X(x_{s_1}, x_{s_2}, \dots, x_{s_n}) \quad (2)$$

where $\{1, 2, \dots, n\}$ is any permutation of $\{s_1, s_2, \dots, s_n\}$. Thus, the sequence is considered exchangeable [14]. Based on this, CP ensures that the following relationship holds:

$$P(y_{n+1} \in \Gamma(x_{n+1})|D) \geq 1 - \alpha \quad (3)$$

where Γ is the prediction interval, D is the dataset that satisfies exchangeability, and α is the significance level. This paper adopts an enhanced version of the original CP method, namely Inductive Conformal Prediction (ICP), to improve the computational efficiency of the CP method. In this framework, the dataset $D = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ is divided into a training set $D_{\text{train}} = \{(X_1, Y_1), \dots, (X_m, Y_m)\}$ and a calibration set $D_{\text{cal}} = \{(X_{m+1}, Y_{m+1}), \dots, (X_n, Y_n)\}$. First, a basic prediction model $\hat{f}$ is trained using the training set D_{train} . Then, a non-conformity score is calculated for each sample in the calibration set D_{cal} . The non-conformity score measures the deviation of calibration set samples from those in the training set. For a data point (X_i, Y_i) in the calibration set, the non-conformity score is defined as:

$$s_i = A(\hat{f}(\mathbf{X}_i), Y_i), \mathbf{X}_i \in D_{\text{cal}} \quad (4)$$

where $\hat{f}(\mathbf{X}_i)$ is the predicted output of the base prediction model for $\mathbf{X}_i$, and Y_i is the corresponding true target value. As the non-conformity score function, which measures the deviation between the predicted value $\hat{f}(\mathbf{X}_i)$ and the actual value Y_i , commonly using absolute error. To construct variable-width prediction intervals, this paper introduces Normalized Conformal Prediction (NCP), where the non-conformity score calculation is as follows [19]:

$$s_i = \frac{|Y_i - \hat{f}(\mathbf{X}_i)|}{\exp(\mu_i)}, \mathbf{X}_i \in D_{\text{cal}} \quad (5)$$

where, μ_i is the predicted value of $\ln(|Y_i - \hat{f}(\mathbf{X}_i)|)$ based on a regression model trained on the training dataset. The NCP

constructs a prediction interval for the test point $(\mathbf{X}_{n+1}, Y_{n+1})$ as:

$$\Gamma(\mathbf{X}_{n+1}) = [\hat{f}(\mathbf{X}_{n+1}) - s_{\delta} \exp(\mu_{n+1}), \hat{f}(\mathbf{X}_{n+1}) + s_{\delta} \exp(\mu_{n+1})] \quad (6)$$

where s_{δ} is the value at position δ of the non-conformity score from the calibration set, sorted in descending order, with $\delta = \lfloor \alpha(n - m + 1) \rfloor$, the symbol $\lfloor \cdot \rfloor$ denotes the floor function.

Additionally, CP provides a significant metric, p-value, which is calculated based on the non-conformity score to assess the deviation of a new data point from the previous dataset. The p-value for the test point $(\mathbf{X}_{n+1}, Y_{n+1})$ is calculated as follows:

$$p_{n+1} = \frac{|i : s_i > s_{n+1}| + \theta |i : s_i = s_{n+1}|}{n - m}, s_i \in D_{\text{cal}} \quad (7)$$

constructing an augmented training set. This augmented training set is then used to further train the model, aiming to enhance the accuracy of the model and effectively reduce data labeling costs. The non-conformity scores, along with the corresponding p-values and martingale values, for the calibration and test sets are calculated based on the updated model. By establishing an appropriate threshold, any point at which the martingale value surpasses this threshold is identified as a location where CD occurs. Subsequently, following the detection of CD, the model is retrained with new data to adapt effectively to the changes introduced by the CD.

3.1. CP-based semi-supervised learning method

This paper introduces an effective method for identifying high-reliability samples within the CP framework. CP quantifies the uncertainty of a sample by providing prediction intervals at specified confidence levels. Within this framework, the NCP assesses the uncertainty of a sample based on the width of its interval. Specifically, the narrower interval at a given confidence level indicates reduced uncertainty in the model's predictive ability. Consequently, this paper focuses on selecting samples with the narrowest intervals to create a high-reliability sample set. This set is then integrated with the existing labeled training dataset to develop an enhanced training dataset. The comprehensive steps of this method are outlined in Algorithm 1.

Algorithm 1. : Semi-supervised learning method based on CP framework

Input: Dataset $D = (\mathbf{X}, Y)$, where $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^n$, D is divided into training set $D_{\text{train}} = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_m, Y_m)\}$ and calibration set $D_{\text{cal}} = \{(\mathbf{X}_{m+1}, Y_{m+1}), \dots, (\mathbf{X}_n, Y_n)\}$, training set D_{train} contains labeled data D_L and unlabeled data D_U , regression model f , test set $T_{\text{data}} = \{(\mathbf{X}_{n+1}, Y_{n+1}), \dots, (\mathbf{X}_{n+q}, Y_{n+q})\}, q \geq 1$, an interval width threshold δ .

Output: Retrained model f_{new} .

Step 1: Normalization: Normalize D_L , D_{cal} , D_U and T_{data} ;

Step 2: Model Training: Fit the regression model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ using the labeled data D_L ;

Step 3: Prediction on Unlabeled Data: Predict the unlabeled dataset D_U and calculate its non-conformity score according to Eq. (5);

Step 4: Interval Prediction: Calculate the corresponding prediction intervals and their widths for each unlabeled data point according to Eq. (6);

Step 5: Data Filtering: Filter out data points whose interval width is less than the threshold value δ adding them to the high-reliability sample set $U_{\text{high_conf}}$;

Step 6: Data Merging: Combine $U_{\text{high_conf}}$ with the original labeled training set D_L to create an extended training set $D_{\text{train_expanded}} = \{D_L, U_{\text{high_conf}}\}$;

Step 7: Model Retraining: Retrain the regression model using $D_{\text{train_expanded}}$ to obtain the updated model f_{new} .

where θ is a random number in the interval $[0, 1]$ and $n - m$ is the number of data points in the calibration set.

3. The proposed algorithm

This paper proposes a novel semi-supervised CD detection method based on the CM framework, as illustrated in Fig. 3. Specifically, the dataset is divided into a training set, a calibration set, and a test set, with the training set comprising labeled and unlabeled datasets in approximately a 1:3 ratio. A regression model is trained using the labeled training dataset, and the prediction intervals for the unlabeled data are computed using NCP. Subsequently, samples with narrower prediction intervals are selected for inclusion in the high-reliability sample set, thus

Remark 1. : This algorithm leverages the CP framework to quantitatively assess the uncertainty of data points using confidence intervals. It specifically selects samples from the unlabeled data with low uncertainty, identified by their prediction interval widths falling below a predetermined threshold. This selection mechanism ensures that only samples exhibiting high predictive reliability are incorporated into the training set. These high-reliability samples are subsequently utilized alongside the labeled data to retrain the regression model, aiming to enhance the model accuracy. In practice, the selection of the confidence level and the interval width threshold, as well as the initial tuning of the regression model, should be optimized and adjusted based on the specific data characteristics and the application requirements.

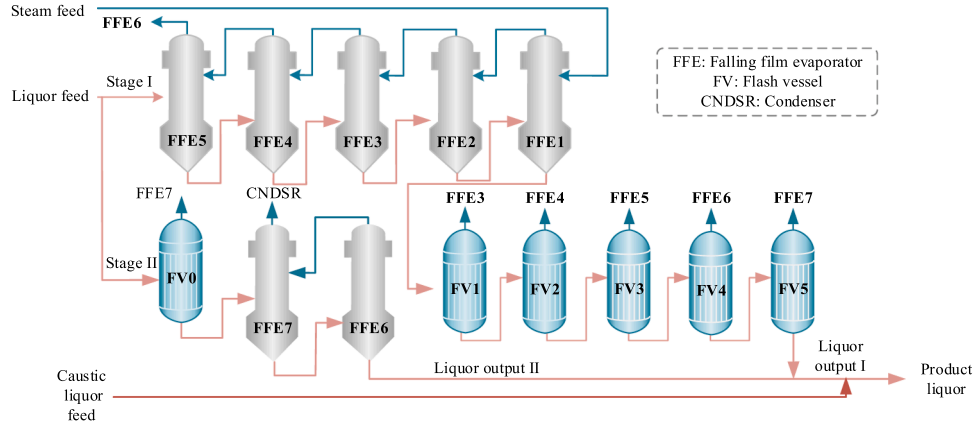


Fig. 4. Alumina backward-feed seven-effect evaporation process.

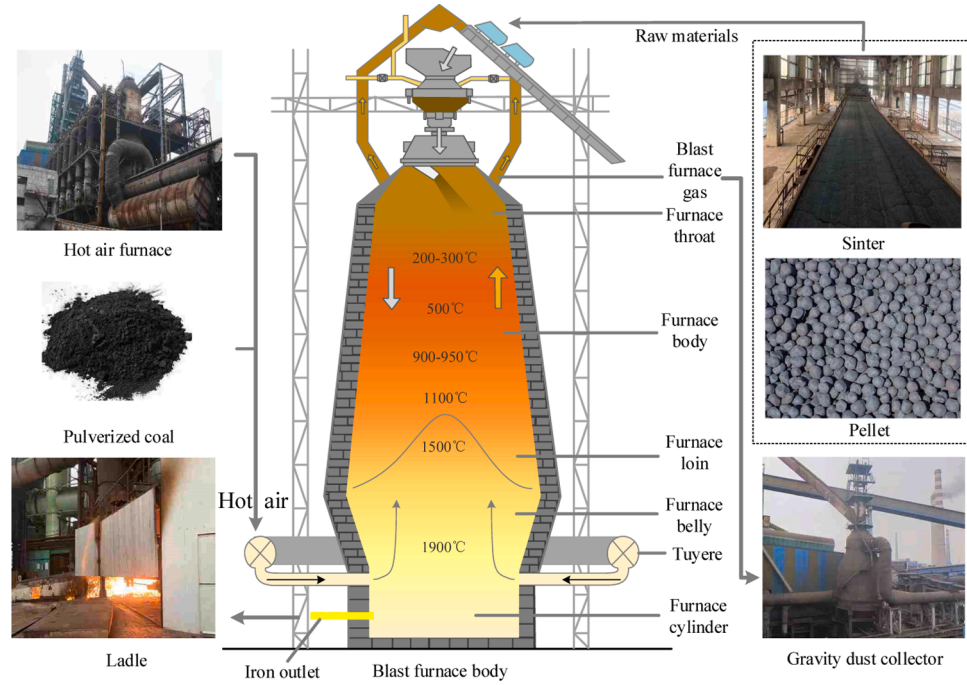


Fig. 5. Schematic diagram of blast furnace ironmaking process.

3.2. CM-based CD detection method

CM is a statistical technique that integrates CP with martingale theory. The martingale is a mathematical tool used for modeling a fair game process, where the expected value of the next observation is equal to the current observation given the past information. CM is constructed from a sequence of values generated by CP. Under the exchangeability assumption, the sequence of p-values follows a uniform distribution, leading the CM sequence to exhibit a stabilizing trend. However, when CD occurs, the data no longer satisfy the exchangeability assumption; consequently, the distribution of the value series deviates from uniformity, leading the CM series to exhibit a consistent upward trend over time. The CM value constructed based on the p-value series is calculated as follows:

$$M_n^{(\varepsilon)} = \prod_{i=1}^n (ep_i^{\varepsilon-1}) = M_{n-1}^{(\varepsilon)} \cdot \varepsilon \cdot p_n^{\varepsilon-1} \quad (8)$$

where, $\varepsilon \in (0, 1)$, $M_n^{(\varepsilon)}$ represents the CM value of the n -th sample point, $M_{n-1}^{(\varepsilon)}$ denotes the CM value of the $(n-1)$ -th sample point, and p_n denotes the p-value of the n -th sample point.

Based on the principle described, when the monitored CM value exceeds a predetermined threshold, this indicates a violation of the exchangeability assumption, thus effectively detecting the occurrence of CD. The specific steps for implementing CD detection based on the CM framework are detailed in [Algorithm 2](#).

Algorithm 2. : CD detection based on the CM framework

Table 1
Overview of dataset splits for different datasets.

Dataset	Sample size	Labeled set	Calibration set	Unlabeled set	Test set	High-reliability set
UCI	9568	1607	690	5357	1914	3878
AL2O3	4131	654	281	2179	1017	1304
BF	5870	986	423	3287	1174	2302

Input: Dataset $D=(X,Y)$, where $X \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^n$, calibration set D_{cal} , $T_{data} = \{(X_{n+1}, Y_{n+1}), \dots, (X_{n+q}, Y_{n+q})\}, q \geq 1$, regression model f_{new} , concept drift threshold θ .

Output: List of indices where concept drift is detected, denoted as $drift_index=[]$.

Step 1: Initialize the CM sequence: $M[0]=1$;

Step 2: For i in $range(len(T_{data}))$:

Step 3: Calculate the non-conformity score of the calibration set using Eq. (5);

Step 4: Calculate the non-conformity score of the test set using Eq. (5);

Step 5: Calculate the corresponding p-value using Eq. (7);

Step 6: Update the martingale value $M[i]$ according to Eq. (8);

Step 7: If $M[i] > \theta$, concept drift is detected at this point;

Step 8: Append i to $drift_index$;

Step 9: Reset the CM value at this position $M[i]=1$;

Step 10: Update the model f_{new} with new data $(T_{data}[i:], T_{data}[i:])$;

Step 11: End.

Remark 2. : Algorithm 2 employs the CM framework to detect CD in the data stream. Initially, the algorithm utilizes non-conformity scores from both the calibration and test sets, combining them to compute a p-value for each data point. By continuously updating the martingale value, the method dynamically monitors and identifies the occurrence of CD. Once the martingale value exceeds a preset threshold θ , CD is considered to have occurred at that location and is recorded. In addition, each time a drift is detected, the algorithm resets the martingale value and updates the regression model with new data to accommodate the

latest changes in the data stream. This approach is particularly well-suited for application scenarios that require real-time responses and adaptation to dynamically changing environments, thereby significantly enhancing model adaptability and prediction accuracy.

4. Experiments and analysis

This paper concentrates on semi-supervised CD detection in regression tasks and validates the effectiveness of the proposed method

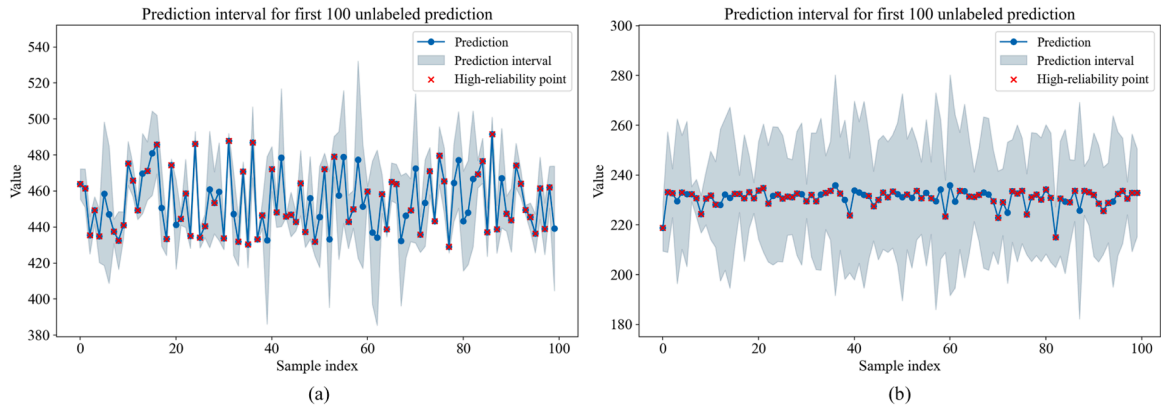


Fig. 6. UCI dataset (a) and alumina evaporation process dataset (b) high-reliability sample selection (first 100 samples as an example).

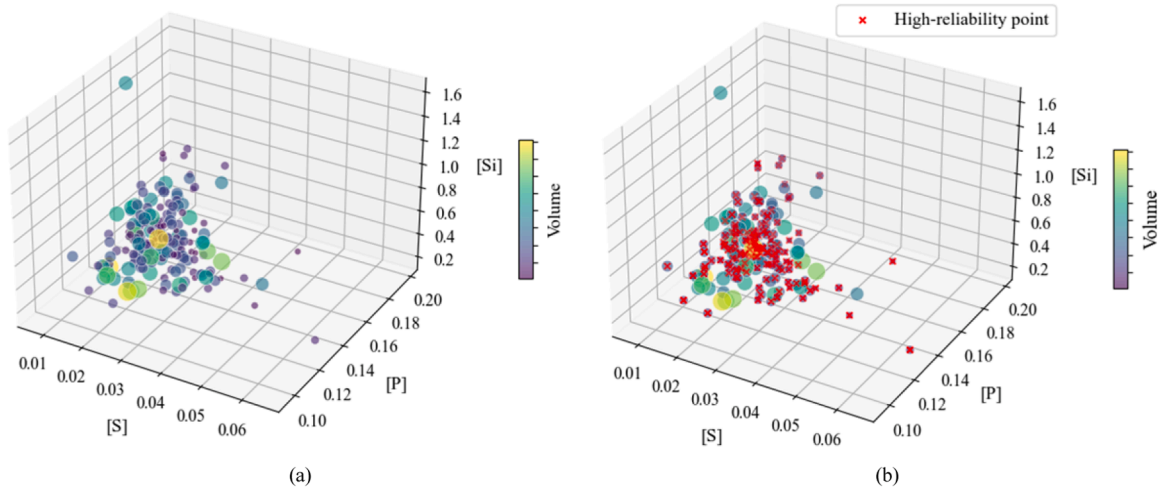


Fig. 7. Three-dimensional diagram of high reliability sample selection for the blast furnace ironmaking dataset (example of the first 200 samples, where darker colors indicate higher reliability and lighter colors indicate lower reliability).

Table 2
Detailed performance indicators for different datasets.

Metric	Dataset	Training set	High-reliability training set	Test set	Post-detection test set
MAE	UCI	1.3120	1.1281	3.1068	0.8880
	AL2O3	7.2488	6.7515	8.0763	3.6705
	BF	0.0239	0.0186	0.0381	0.0151
MSE	UCI	3.0946	2.4311	16.3225	1.5985
	AL2O3	66.9784	53.0743	89.2823	30.0896
	BF	0.0013	0.0009	0.0048	0.0011

through a series of experiments and analysis. Initially, the proposed method is validated using the UCI dataset to confirm its effectiveness. Subsequently, to further explore the practice value of the proposed method in real-world scenarios, additional experiments are conducted

on datasets from the alumina evaporation process and the blast furnace ironmaking process. These experiments not only validate the effectiveness of the methodology in this paper, but also demonstrate its potential applicability in various industrial processes. Moreover, the mean absolute error (MAE) and mean square error (MSE) are employed as the evaluation metrics to assess the performance of the approach.

4.1. Experimental data introduction

UCI dataset: A dataset is derived from the combined cycle power plant dataset provided by the UCI Machine Learning Repository [20], consisting of 9568 data points involving four key input variables: Ambient Temperature (AT), Atmospheric Pressure (AP), Relative Humidity (RH), and Vacuum (V); the output variable is Full Load Electrical Power Output (P_E).

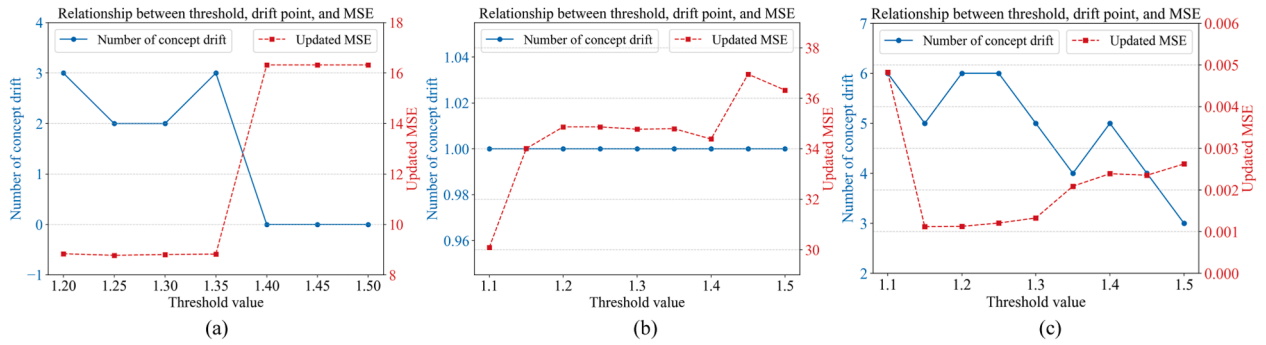


Fig. 8. Performance across different thresholds on the UCI dataset (a), the alumina evaporation process dataset (b), and the blast furnace ironmaking dataset (c).

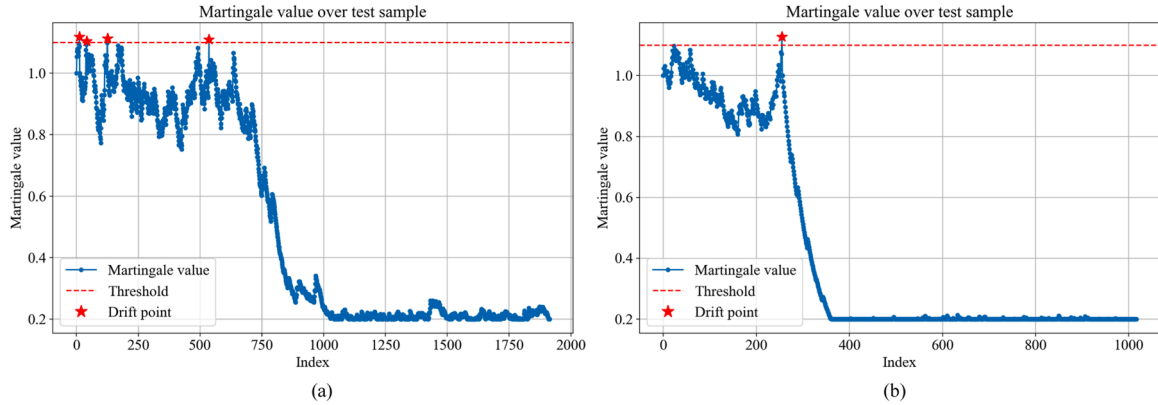


Fig. 9. Concept drift detection map based on UCI dataset(a) and alumina evaporation process dataset(b).

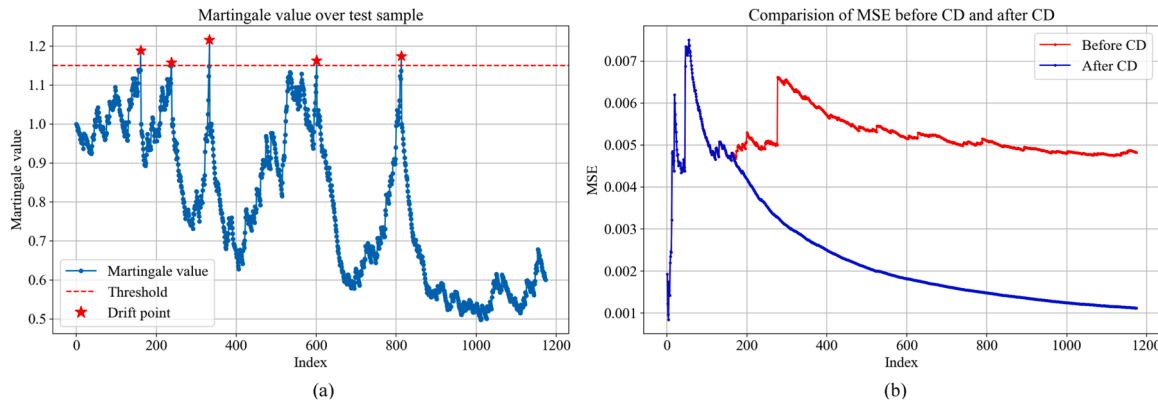


Fig. 10. Concept drift detection map based on blast furnace ironmaking dataset(a) and comparison of MSE before and after detection based on blast furnace ironmaking dataset(b).

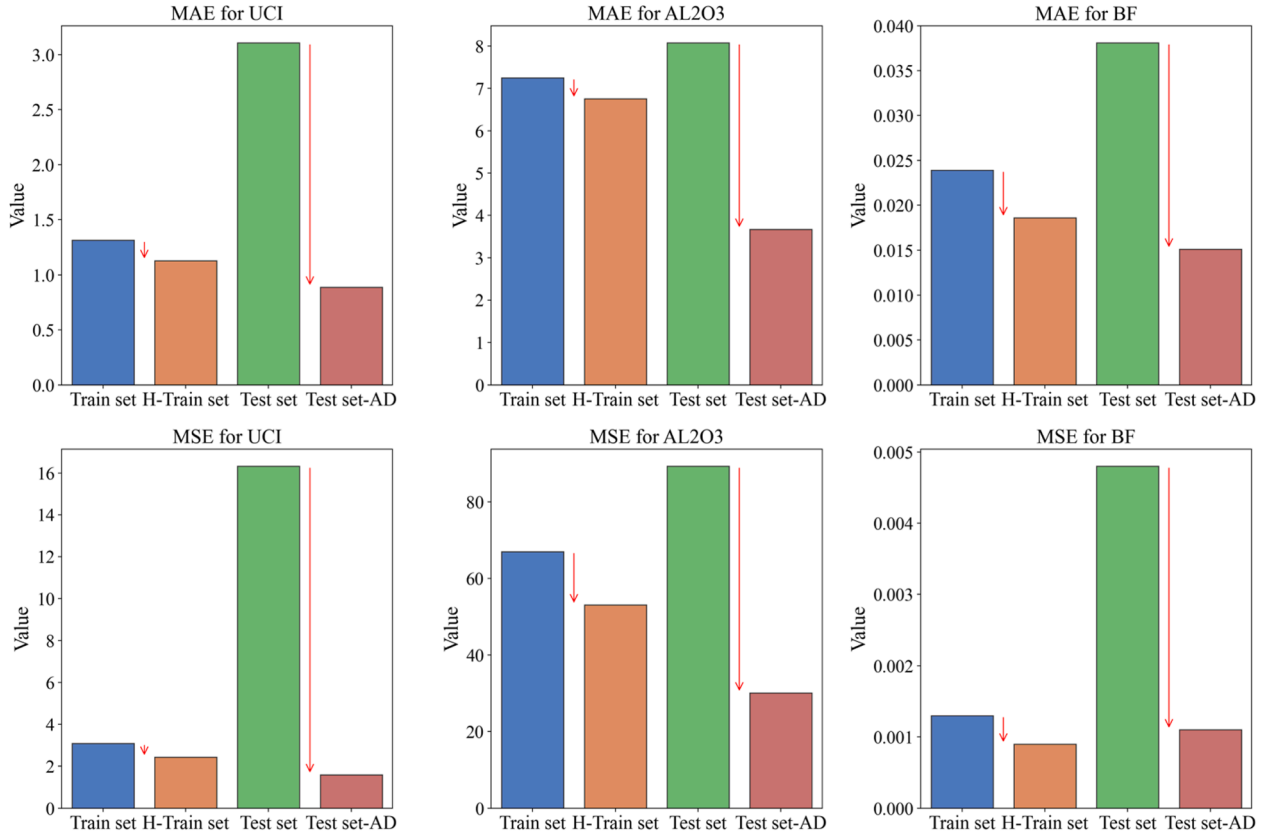


Fig. 11. Comparison of MAE and MSE of semi-supervised and concept drift detection methods with different datasets (where H-Train Set stands for High Confidence Training Set, Test Set-AD stands for Test Set After Detection).

Table 3

Comparison of CD detection points and updated MSE results across various detection methods on the test set.

Method	Concept drift point	Updated MSE
ADWIN[11]	{1791}	15.4439
PageHinkley[6]	{None}	16.3225
KSWIN[25]	{742, 906}	7.2795
The proposed method	{12, 41, 125, 536}	1.5985

Alumina evaporation process dataset: This dataset is originated from a backward-feed seven-effect evaporation process in an alumina plant in Shanxi. The detailed flow of this evaporation process is illustrated in Fig. 4. The evaporation process is a critically important component of the Bayer method for alumina production. The primary purpose of this process is to concentrate an aqueous solution by evaporation, controlling the alkali concentration and inducing the precipitation of impurity solutes in the evaporator to remove impurities [21].

Blast furnace ironmaking dataset: This dataset is derived from the operational data of Blast Furnace No. 2, located in a region of Southwest China. The ironmaking process of this furnace is illustrated in Fig. 5. The blast furnace ironmaking process is a critical component of steel production, with production efficiency and iron quality being crucial to the entire process. Real-time monitoring of the blast furnace operating status is essential to maintain the continuity and efficiency of production. Traditionally, the contents of Silicon ([Si]), Sulfur ([S]), and Phosphorus ([P]) are considered crucial indicators for assessing the operating condition of the blast furnace [22–24]. However, these parameters are challenging to measure using online methods. This limitation necessitates the maintenance of highly accurate predictive models to effectively monitor the condition of blast furnace. The dataset contains 16 key process variables affecting the quality of the molten iron as input

variables, with the contents of [Si], [S] and [P] serving as the output variables.

To tackle the challenges of real-time steaming data, this study employs an incremental regression model based on XGBoost, which is initially trained and subsequently updated in response to CD detection. During the data preprocessing stage, each dataset is divided into training, calibration, and test sets in an approximate ratio of 7:1:2. Within the training

set, the ratio of labeled to unlabeled data is about 1:3. In addition, details of the sample size distribution for each dataset are shown in Table 1.

4.2. Analysis of semi-supervised experimental results based on CP framework

High-reliability samples from various datasets are identified using the CP framework, with portions of the screening process are illustrated in Fig. 6 and Fig. 7. For the UCI dataset and alumina evaporation process datasets, samples are chosen for the high-reliability set based on narrower prediction intervals. In the case of the blast furnace ironmaking dataset, given that the output contains three indicators, these indicators are treated as dimensions in a three-dimensional space. The product of these dimensions represents the “volume” of the samples. Samples with smaller volumes, represented by darker points in Fig. 7, are selected as the high-reliability samples. By establishing specific thresholds, approximately 70 % of the unlabeled data and their pseudo-labels are filtered out from the UCI, alumina evaporation process, and blast furnace ironmaking datasets to construct the high-reliability samples. The specific values are presented in Table 1.

Within the semi-supervised learning framework, the process of screening high-reliability samples using the CP method significantly enhances the quality of pseudo-labels and effectively minimizes the

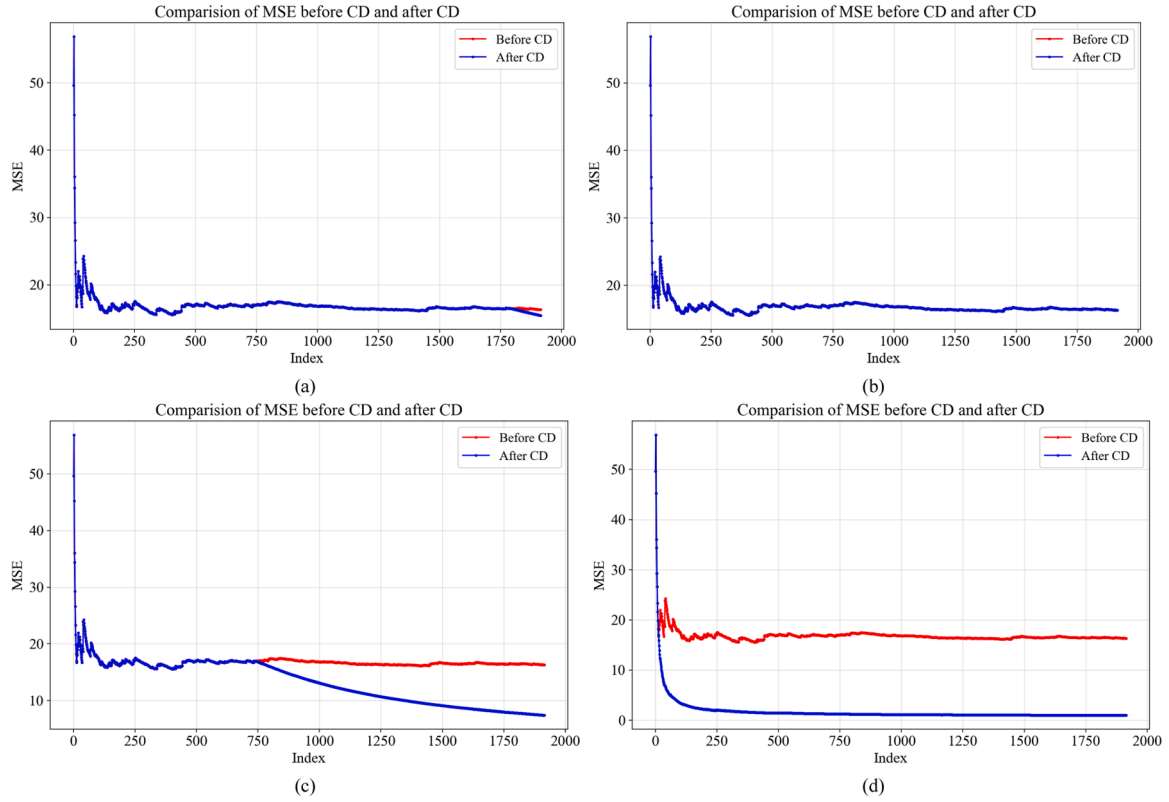


Fig. 12. Trend of MSE updates across different concept drift detection methods on the UCI test set: (a) ADWIN, (b) PageHinkley, (c) KSWIN, and (d) the proposed method.

interference from uncertainty samples during model training, thereby boosting the prediction accuracy of the model. As demonstrated in Table 2, the MAE for the UCI dataset drops from 1.3120 to 1.1281, while the MSE reduces from 3.0946 to 2.4311. Similarly, in the alumina evaporation process dataset, the MAE falls from 7.2488 to 6.7515, and the MSE from 66.9784 to 53.0743. In the blast furnace iron-smelting dataset, the MAE decreases from 0.0239 to 0.0186, and the MSE from 0.0013 to 0.0009. Fig. 10 vividly illustrates these reductions in metrics, with the MAE decreasing by 14.02 % and the MSE by 21.44 % in the UCI dataset; the MAE by 6.86 % and the MSE by 20.76 % in the alumina evaporation process dataset; and in the blast furnace ironmaking dataset, the MAE by 22.18 % and the MSE by 30.77 %. These results demonstrate that the use of the CP framework in semi-supervised learning reduces the uncertainty of the model when dealing with unlabeled data, enhancing the overall performance of the model and leading to more stable and accurate predictions across complex industrial datasets.

4.3. Analysis of experimental results for CD detection based on the CM framework

Within the CD detection process utilizing the CM framework, a data point is identified as a CD occurrence whenever the CM value surpasses a predetermined threshold. This study conducts a sensitivity analysis on the thresholds for CD detection across various datasets, as depicted in Fig. 8. To circumvent the establishment of excessively low thresholds, which could precipitate the superfluous detection of drift points and thereby squander computational resources, the study considers thresholds ranging from 1.1 to 1.5. The evaluation metrics include the number of detected CD points and the MSE following model updates. Regarding the UCI dataset, as illustrated in Fig. 8(a), a threshold of 1.1 is deemed optimal due to its substantial reduction in MSE after model upgrades. For the alumina evaporation process dataset, depicted in Fig. 8(b),

although the number of detected drift points remains constant across all thresholds, the threshold of 1.1 is selected as the most favorable, correlating with the minimal updated MSE. In the case of the blast furnace ironmaking dataset, shown in Fig. 8(c), a threshold of 1.15 is determined to be optimal, corresponding to the lowest updated MSE.

The CD detection results, as illustrated in Fig. 9 and Fig. 10(a), four significant drift points are identified in the UCI dataset, one in the alumina evaporation process dataset, and five in the blast furnace ironmaking dataset. The presence of these drift points signifies a notable change in data distribution, necessitating model adjustments to preserve prediction accuracy. During the experiments, the model undergoes retraining using new data following the detection of drift points.

Table 2 showcases the test set error performance both before and after the detection of CD. Specifically, in the UCI dataset, the MAE decreases from 3.1068 to 0.8880, making a reduction of approximately 71.42 %, and the MSE drops from 16.3225 to 1.5985, a reduction of roughly 90.21 %; in the alumina evaporation process dataset, the MAE falls sharply from 8.0763 to 3.6705, a substantial reduction of approximately 54.55 %, and the MSE plummets from 89.2823 to 30.0896, a decrease of roughly 66.30 %; in the blast furnace ironmaking dataset, the MAE decreases from 0.0381 to 0.0151, reflecting a decrease of about 60.37 %, and MSE decreases from 0.0048 to 0.0011, showing a reduction of approximately 77.08 %. Fig. 10(b) illustrates the MSE trend for the test set before and after CD detection, clearly showing that the prediction error of the model increases significantly at the first drift point, while the performance continues to improve after retraining. Fig. 11 further demonstrates the significant reduction in the overall MAE and MSE across each test set following CD detection and model retraining. This confirms that the method proposed in this paper not only effectively addresses the performance degradation caused by CD but also provides innovative tools and insights for maintaining the model's performance in dynamic industrial settings. This strategy enhances the model's adaptability to potential future changes, making it

particularly well-suited for industrial applications where the environment and operating conditions are subject to frequent variations.

4.4. Comparative experiments on different CD detection methods

To underscore the superiority of the CM-based CD detection method proposed in this paper, comparative experiments are conducted against three widely adopted baseline methods for regression tasks: ADWIN [11], PageHinkley [6], and

KSWIN [25]. These experiments utilize datasets from the UCI repository. Table 3 summarizes the number of detected drift points and the corresponding updated MSE values for each method. The results indicate that the proposed method achieves superior performance in terms of the updated MSE metric. While the proposed method identifies a slightly higher number of drift points, this trade-off is acceptable given the substantial reduction in MSE.

Fig. 12 illustrates the variations in MSE on the test set before and after model updates across the various CD detection methods. As depicted in the figure, the ADWIN method detects only one drift point, leading to a minimal reduction in MSE. The PageHinkley method fails to detect any drift points, resulting in no improvement in MSE. Although the KSWIN method detects two drift points and achieves a moderate reduction in MSE, the improvement remains limited. In contrast, the proposed CM-based method significantly reduces MSE to the lowest level, demonstrating a pronounced and substantial advantage over the baseline methods.

5. Conclusion

In this study, a novel semi-supervised CD detection methodology based on the CM framework is introduced, addressing the CD phenomenon that is widespread in a numerous industrial data streams. The approach is specifically designed to confront the dual challenges of high cost of data labeling and effective CD detection. The proposed method significantly enhances the quality of pseudo-labeling and drastically diminishes the model prediction error through a precise CD detection and retraining mechanism. The efficacy of the approach has been demonstrated through multiple regression tasks in various real-world industrial contexts. Moreover, the adaptability and utility of the proposed framework are not confined to regression tasks; they also expand to encompass other data analysis tasks such as classification and clustering. Overall, the research presents a robust and flexible tool for analyzing dynamically changing industrial data streams, ensuring the stability and precision of the model in environments subject to frequent fluctuations.

CRedit authorship contribution statement

Yu Zhang: Investigation, Methodology, Validation, Writing—original draft, Software. **Ping Zhou:** Conceptualization, Methodology, Writing—review & editing, Formal analysis, Supervision. **Ruiyao Zhang:** Methodology, Writing—review & editing. **Shaowen Lu:** Writing—review & editing, Data curation. **Tianyou Chai:** Formal analysis, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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